INCH-POUND

MIL-M-44479
31 March 1994

MILITARY SPECIFICATION

MEATBALLS IN BROWN GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers meatballs in brown gravy, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations (9 CFR Parts 301-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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U.S. Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Grades of Tomato Paste (7 CFR Part 52.5041)

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0709, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

USDA Northern Regional Research Laboratory Odor and Flavor

Intensity Scoring Scale of the Cereal Panel Odor and Flavor Evaluation System

(Copies are available from the USDA Northern Laboratory Research Center (ARS-USDA), 1815 North University, Peoria, IL 61604.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)
U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

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(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. Beef shall be from steers, heifers, or cows and shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specification (IMPS) Cuts: Item No. 114 (chuck, shoulder clod), 115 (chuck, square cut, boneless), 128 (chuck, crosscut, boneless), 133 (triangle, boneless), 161 (round, shank off, boneless), 163

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(round, shank off, 3-way, boneless), 165 (round, rump end, shank off, boneless), 167A (knuckles, peeled), 169 (top round), 170A (bottom round, heel out), and 171B (outside round. All beef shall be certified by a USDA, Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the IMPS Quality Assurance Provisions.

NOTE: Certified IMPS Cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification, as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled as "Beef for further processing as specified in MIL-M-44479". The certificate shall be noted as such and shall accompany the shipment.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef, prior to processing into the finished product, shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained at an internal temperature of 28°F to 50°F, inclusive.

b. Holding in the fresh-chilled state for not more than 4 days after initial certification is permitted, provided that the beef is maintained at an internal temperature of 28°F to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence

d. Tempering/thawing of the frozen beef shall comply with the following:

- microwave tempering/thawing of beef is permitted. Any evidence of cooking, scorching, or browning caused by the microwave process shall be cause for rejection of the product which cannot be reworked or reoffered. Additionally, the product shall show no evidence of deterioration due to mishandling prior to thawing (for example, discoloration, foreign odor, excessive purge, or obvious deviation from specification handling and storage requirements) as determined by USDA
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation

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- regardless of the process used, frozen beef shall be tempered/thawed to an internal temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- beef which has been tempered/thawed shall not be refrozen.

3.2.2 Meatballs, cooked, individually quick frozen (IQF). The meatballs shall have been prepared from beef meeting the requirements of 3.2.1 and 3.2.1.1. The meatballs shall have been processed in accordance with 3.3.1 and 3.3.2. The cooked IQF meatballs shall be packaged and sealed in USDA approved food packaging material. The cooked, packaged meatballs shall have been held at an internal temperature of 0°F or below for not more than 90 days prior to tray pack filling. IQF cooked meatballs shipped between plants shall be accompanied by a USDA AMS Certificate to certify the initial freezing (in-storage) date and compliance with the above stated requirements.

3.2.3 Bread, crumbs. Bread crumbs shall be produced from fresh, white, yeast raised bread. The shortening used in the bread making shall be of vegetable origin only and shall contain BHA at a level not in excess of 0.02 percent of the fat content of the bread. The moisture content of the bread crumbs shall not exceed 8.0 percent. The bread crumbs shall be coarse ground, free-flowing and shall be free from burnt, rancid, moldy or musty odors and flavors.

3.2.4 Color, caramel, powder. Powdered caramel shall comply with the Food Chemicals Codex (see 6.4.1).

3.2.5 Flavoring, natural, roast beef. Natural roast beef flavor with other natural flavors shall consist of approximately seventy five percent rare roast beef and approximately twenty five percent braised beef flavors. The color shall be a tan to light brown, free-flowing powder. A one percent water solution shall demonstrate typical roast beef and braised beef flavor and aroma notes. The moisture content of the powder shall not exceed 8.0 percent (see 6.4.2).

3.2.6 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.7 Lecithin, powder. Powdered lecithin shall comply with the Food Chemicals Codex.

3.2.8 Onions, chopped, dehydrated. Dehydrated chopped onions shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

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3.2.9 Onion powder. Onion powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.10 Pepper, black, ground. Ground black pepper shall be derived from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground black pepper shall have a characteristic penetrating odor, a hot, biting, pungent flavor, and a light gray to speckled black-gray color. The ground pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground black pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.11 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.2.12 Protein, soy, concentrate. The soy protein shall be a textured soy concentrate and shall comply with the following (see 6.4.3):

1. Chemical Characteristics:

a.	Protein (N x 6.25) moisture free basis	70.0% (min)
b.	Moisture	8.8% (max)
c.	Fiber, crude	4.5% (max)
d.	Ash	6.0% (max)
e.	Fat (American Oil Chemists Society Method BE 3-49)	1.0% (max)
f.	Caramel color, double strength	0.2 to 1.0% 1/

1/ Twice the amount of single strength may be used to meet this requirement.

2. Flavor Characteristics:

a. Shall receive a flavor score of 6.0 or higher when tested in accordance with the provisions of the USDA Northern Regional Research Laboratory Odor and Flavor Intensity Scoring Scale of the Cereal Panel Odor and Flavor Evaluation System.

b. Shall have no artificial flavor added.

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3.2.13 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.14 Starch, modified, high opacity. High opacity starch shall be white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The modified starch shall resist breakdown at low pH and under shear stress. A cooked starch slurry shall be bland with essentially no cereal or starch taste.

3.2.15 Tomato paste. Tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste.

3.2.16 Water. Water used for formulation, blanching, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef meeting the requirements of 3.2.1 shall be further trimmed, if necessary, to ensure compliance with the finished product requirements. The beef shall be coarse ground through a plate with holes measuring 1 inch in diameter. The ground beef shall be held for not more than 24 hours at an internal temperature of 28°F to 40°F prior to meatball preparation.

3.3.2 Meatball preparation. The meatballs shall be formulated and prepared as follows:

Ingredient	Percent by weight
Beef, ground	89.58
Bread crumbs	4.25
Water	2.85
Onions, chopped, dehydrated	1.25
Protein, soy, concentrate	1.00
Salt 1/	0.90
Pepper, black, ground	0.09
Garlic powder	0.08

1/ The total amount of salt in the formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

a. The ingredients shall be combined and thoroughly mixed. The meatball mixture shall be ground through a plate having holes 3/16 inch in diameter or chopped to give equivalent results and formed into meatballs of not less than 1.90 ounces nor more than 2.20 ounces each. The meatball

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mixture and the meatballs shall be maintained at an internal temperature of 28°F to 40°F for not more than 12 hours from time of forming to the browning (cooking) process.

b. Meatballs shall be partially cooked (surface browned) to produce a medium brown color, and to comply with the finished product requirements. The meatballs may be cooked by any commercial method except moist heat (steaming) and deep fat frying. Alternatively, fresh formed meatballs may be placed in boiling water only for time sufficient to firm up and shape the meatballs. Meatballs prepared by the alternate method shall then be partially cooked (surface browned) as specified.

c. Cooked meatballs shall be maintained at an internal temperature of 28°F to 40°F for not more than 48 hours from time of cooking to beginning of the IQF process (see 3.2.2) or to tray pack can filling. Cooling of the meatballs is not necessary if can filling occurs immediately after cooking.

3.3.3 Gravy preparation. The gravy shall be formulated and prepared as follows:

Ingredient	Percent by weight
Water	86.73
Starch, modified, high opacity 1/	4.75
Tomato paste (26 percent solids) 2/	4.00
Onion powder	2.50
Salt 3/	0.90
Flavoring, natural, beef roast	0.65
Lecithin powder	0.20
Color, caramel, powder	0.10
Pepper, black, ground	0.09
Garlic powder	0.08

1/ The total amount of starch in the gravy formula shall be adjusted, as necessary, to produce a product that complies with the finished product viscosity requirements (see 3.6).

2/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solids level to that specified and the water shall be adjusted accordingly.

3/ The total amount of salt in the gravy formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

NOTE: The following gravy preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared gravy are still applicable).

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a. A thin slurry shall be made with the starch and part of the water.

b. The balance of the water and the remaining ingredients shall be mixed together and heated to 140°F to 180°F to form an emulsion.

c. The steam supply to the kettle shall be shut off and the starch slurry shall be added to the emulsion to form the gravy.

d. The volume of the gravy shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The gravy shall be maintained in the temperature range of 150°F to 180°F and filled into tray pack cans within four hours after preparation.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product to conform to the finished product requirements and to the following requirements:

a. A minimum of 36 meatballs shall be placed in a single layer within each tray pack can.

b. The tray pack can of meatballs shall be filled with gravy to a net weight of 106 ounces.

c. Immediately after filling, each can shall be sealed in accordance with the can manufacturer's guidelines/requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable (see 4.5.5), and under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6p).

d. The filled and sealed tray pack cans shall be in the retort process within two hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

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d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual can shall be not greater than 1.0 ounce.

e. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. Each can shall contain not less than 36 intact meatballs.

g. Meatball texture shall not be dry, rubbery, tough, or mushy.

h. Meatballs shall retain their shape.

i. Drained weight of 36 intact meatballs shall be not less than 53.0 ounces.

j. The average drained weight shall be not less than 55.0 ounces.

k. The average net weight shall be not less than 106 ounces.

l. No individual can shall have a net weight of less than 104 ounces.

m. The average fat content of the finished product shall be not greater than 10.0 percent and no individual tray pack shall have a fat content greater than 12.0 percent.

n. No individual tray pack shall have a salt content less than 0.5 percent or greater than 1.2 percent.

o. The product shall show no evidence of excessive heating (materially darkened or scorched).

p. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

q. The gravy shall not be lumpy.

r. The viscosity of the gravy shall be not less than 4.0 cm per 10 seconds nor greater than 12.0 cm per 10 seconds when determined by a Bostwick Consistometer (see 4.5.3.4).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Product Inspection Regulations of the U.S. Department of Agriculture.

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3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the AOAC International, and in the approved methods of the American Association of Cereal Chemists, to determine

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conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight examination. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.5.3.2 Double sampling plan for product examination. The finished product shall be examined for the defects listed in table I utilizing the double sampling plans indicated in MIL-STD-105. The lot size shall be expressed in tray pack cans. The sample unit shall be one filled and sealed tray pack can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for minor defects. The sample cans shall be heated in accordance with heating instructions on the can label.

TABLE I. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in a can is more than 1.0 ounce
102		Presence of a bone piece measuring 0.3 inch or more in any dimension
103		Drained weight of 36 intact meatballs in a can is less than 53.0 ounces 3/ 4/
104		Less than 36 intact meatballs in any individual can

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TABLE I. Product defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
105		Product shows evidence of excessive heating (materially darkened or scorched) 5/
	201	More than one meatball did not retain its shape
	202	Texture of meatballs is dry, rubbery, tough, or mushy
	203	Gravy is lumpy

1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

3/ To determine drained weight, the gravy in each can shall be poured off, strained through a U.S. Standard No. 8 sieve, and individually reserved for viscosity determinations. The remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquefy rendered fat and remove the gravy without breaking the meatballs. The contents shall then be poured into the U.S. Standard 1/2 inch sieve in a manner that will distribute the product over the sieve without breaking the meatballs. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight. Thirty-six meatballs shall be weighed to determine the drained weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ If the sample average drained weight is less than 53.0 ounces, the lot shall be rejected.

5/ Slight scorched marks due to broiling shall not be considered a defect (see 3.3.2).

4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of the

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AOAC, methods 976.21, 985.15, or 960.39, and for salt in accordance with the Official Methods of Analysis of the AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.3.4 Viscosity testing. The strained gravy collected from each of the cans selected for drained weight inspection (see 4.5.3.2 and footnote 3/ to table I) shall be individually tested for viscosity as specified below (see 6.5).

Instrument: Bostwick Consistometer
Catalog number:
23270-004 or 15-347-50
VWR Scientific Company
P.O. Box 7900
San Francisco, CA 94120

Fisher Scientific
585 Alpha Drive
Pittsburgh, PA 15238

Method:

a. Level the instrument.

b. Bring gravy to 100°F ± 1°F in a water bath in a covered container.

c. Stir gravy thoroughly before filling the Bostwick cavity.

d. Scrape gravy evenly across upper edge of cavity.

e. Release gravy and time gravy flow to the nearest one second and measure distance traveled to the nearest 0.1 cm.

If the Bostwick viscosity value of the gravy from any can in the sample fails to conform to the requirement specified in 3.6, it shall be classified as a major defect and the lot shall be rejected.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the U.S. Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, scratches, scuffs, or abrasions that occur on the outside coating as a result of the filling, sealing, and thermoprocessing of the tray pack cans shall not be scored as a defect.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except, for self-adhering labels, the following additional defects shall apply:

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Major: Label torn or scratched so as to obliterate any of the markings

Minor: Air bubbles under label
Label not properly adhered to can, for example, label raised or peeled back from edges or corners

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's guidelines/ requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75°F ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, lay the straight edge only between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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darkened or scorched)

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of class domestic, grade 175 fiberboard.

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5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1994 would be coded as 4079). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 end of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. In addition, cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 to 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2 meatballs each.

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The first sample exceeds the accept- c. If the number of defects found in

As an alternate labeling method, a preprinted, self-adhering 0.002 inch thick clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack can shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Ingredient information.

6.4.1 Color, caramel. It has been found that Dark Caramel Color produced by the Dirigo Corp., meets the requirements of 3.2.4 and performs satisfactorily in this product.

6.4.2 Flavoring, natural, roast beef. It has been found that Natural Roast Beef Flavor WONF (C265-498), produced by Haarmann and Reimer Corp., Springfield, NJ, meets the requirements of 3.2.5 and performs satisfactorily in this product.

6.4.3 Protein, soy, concentrate. It has been found that product number 7181 and Response 4412, produced by Central Soya, Fort Wayne, IN, meets the requirements of 3.2.12 and performs satisfactorily in this product.

6.5 Alternate viscosity testing method. The contracting officer may authorize an alternative contractor-recommended method of viscosity testing if the alternative method is approved by the U.S. Army Natick Research, Development and Engineering Center.

6.6 Subject term (key word) listing.

Canned food
Combat field feeding
Operational ration
Food processing
Shelf stable

Custodians:

Army - GL
Navy - SA
Air Force - 35

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8940-0773)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-44479	2. DOCUMENT DATE (YYMMDD) 1994 March 31

3. DOCUMENT TITLE
MEATBALLS IN BROWN GRAVY, THERMOSTABILIZED, TRAY PACK

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WRE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP4, JUL 91 *Previous editions are obsolete.* 198/290

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